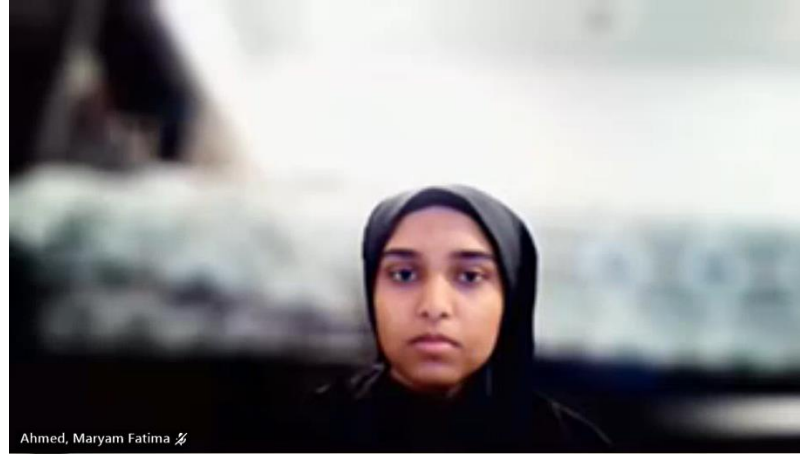


Study Room Booking System

Team 2

Anjali Ayyappan
Cameron Yousif
David Aiyeyemi
Diego Mireles
Maryam Ahmed

Introductions



Objective

The objective of our project is to design and develop a study room booking system that provides students with an easier experience in accessibility of the study spaces on campus.

With our system's platform to reserve study rooms, this will help students to organize their study schedules more efficiently. We hope to maximize the usage of available study spaces with our system, while also minimizing reservation conflicts with students on campus.



Cost Estimation

Using the Function Point Method

- Estimated productivity: 10 FP/person-month
- Estimated development effort: 6.7 person-months
- Estimated total effort: 1,071 hours
- Estimated wage: \$20/hour
- Estimated labor cost: \$21,417

$$\text{GFP} = 3 \times 4 + 4 \times 5 + 3 \times 4 + 2 \times 10 + 1 \times 5 = 69 \text{ FP}$$

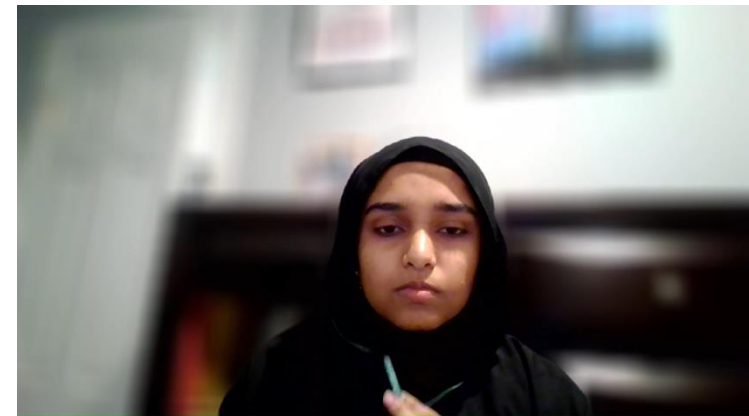
$$\text{PCA} = 0.65 + 0.01 \times (32) = 0.97$$

$$\text{FP} = 69 \times 0.97 = 66.93 \text{ FP.}$$

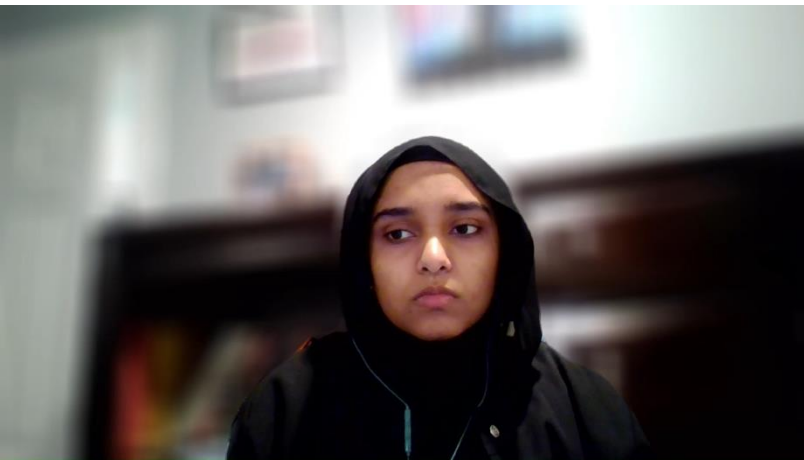
Productivity of 10 FP/person-month

$$E = 66.93 \div 10 = 6.693 \approx 6.7 \text{ person-months or 1,071 hours}$$

$$\text{Labor cost of } 1,070.88 \times \$20 = \$21,417.60.$$



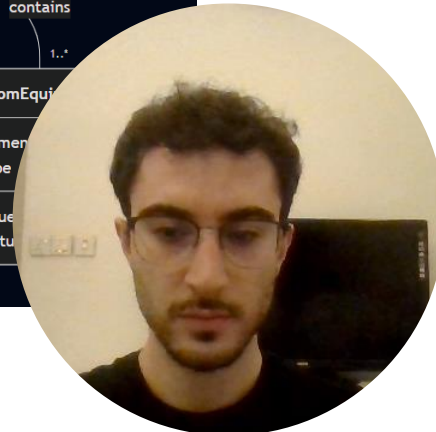
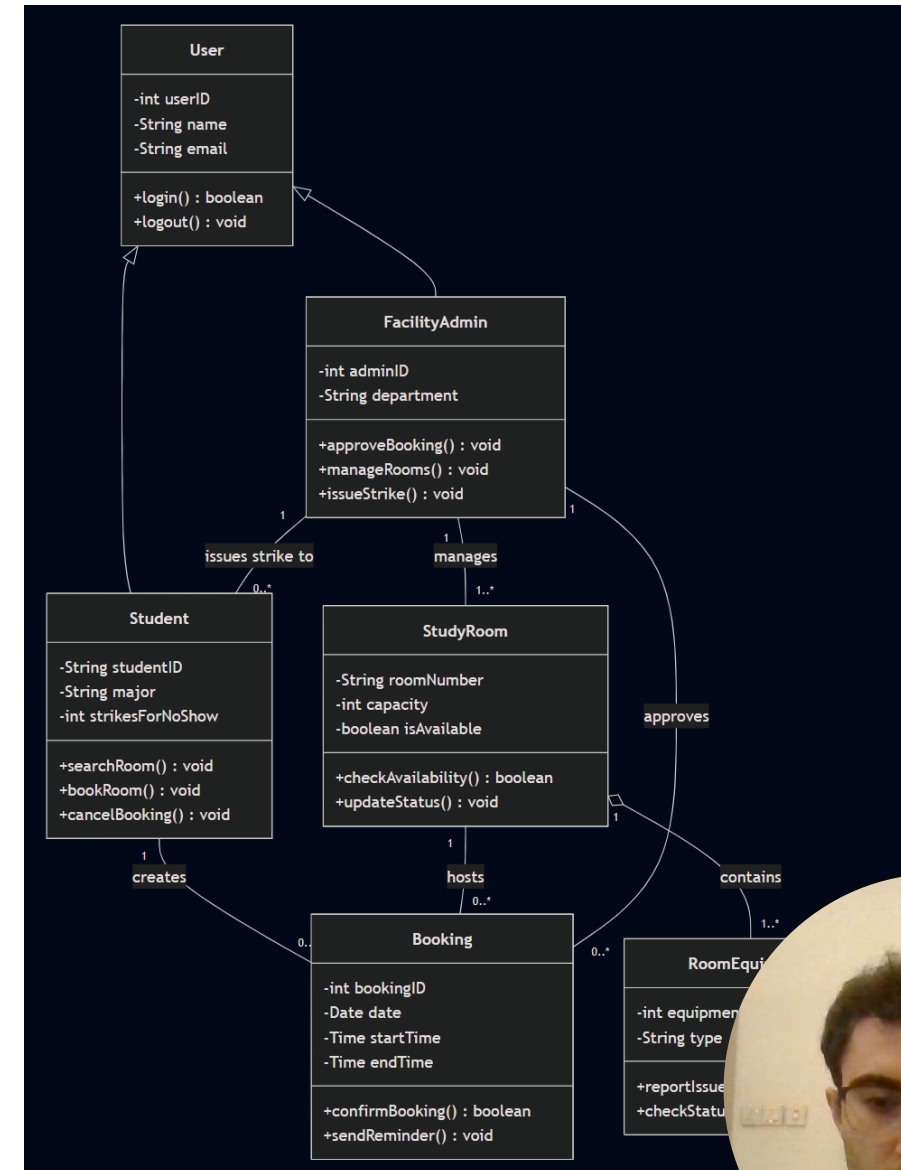
d. Project Timeline



Week	Dates	Main Activities
1	Jun 29 – Jul 3	Requirements & project setup
2	Jul 6 – Jul 10	Database & Spring Boot backend
3	Jul 13 – Jul 17	Authentication & user management
4	Jul 20 – Jul 24	Room searching & availability
5	Jul 27 – Jul 31	Room booking & reservations
6	Aug 3 – Aug 7	Modify/cancel reservations
7	Aug 10 – Aug 14	Admin room management
8	Aug 17 – Aug 21	Frontend/backend integration
9	Aug 24 – Aug 28	Testing & debugging
10	Aug 31 – Sep 4	Bug fixes & refinement
11	Sep 7 – Sep 11	Final testing & documentation
12	Sep 14	Final submission

Class Diagram

- Shows the main classes in our system: User, Student, FacilityAdmin, StudyRoom, Booking, and RoomEquipment
- Student and FacilityAdmin both inherit from User
- A StudyRoom can have many pieces of RoomEquipment
- Students create Bookings, and FacilityAdmins manage StudyRooms and approve Bookings



e. Functional & Non-Functional Requirements

Functional Requirements

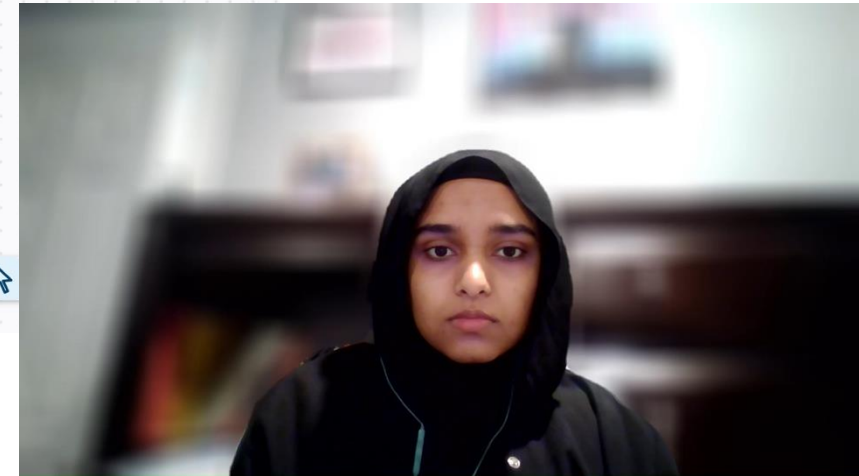
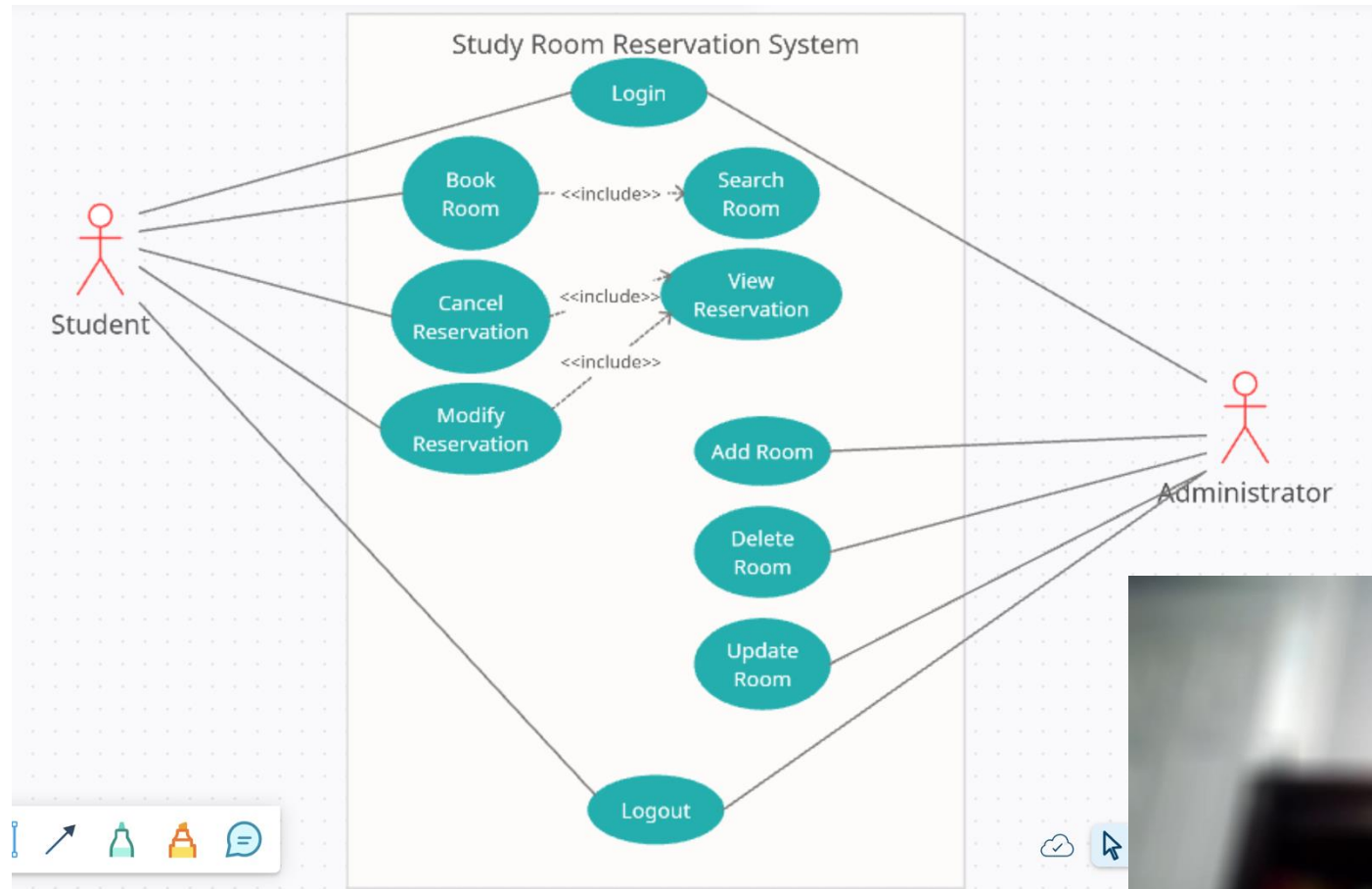
1. The system **shall** allow all users to register their account and authenticate as either a student or an administration.
2. The system **shall** allow all users to view available study rooms based on capacity and equipment.
3. The system **shall** allow all users to add a study room for reservation, including the day of the week in which it is booked as well as the start and end time.
4. The system **shall** allow all users to delete and modify their existing study room reservations.
5. The system **shall** allow all users to view their past and future reservations.

Non- Functional Requirements

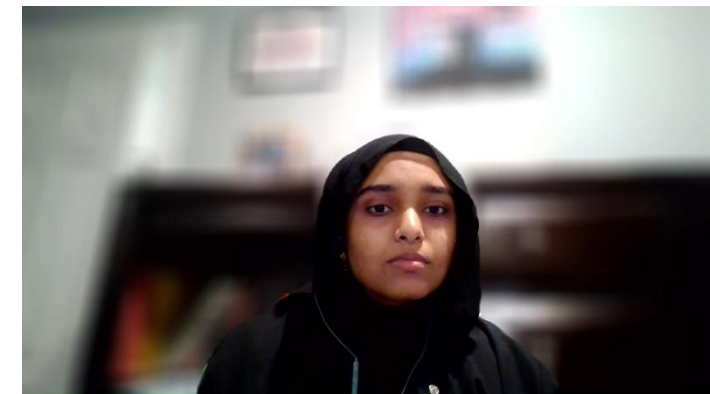
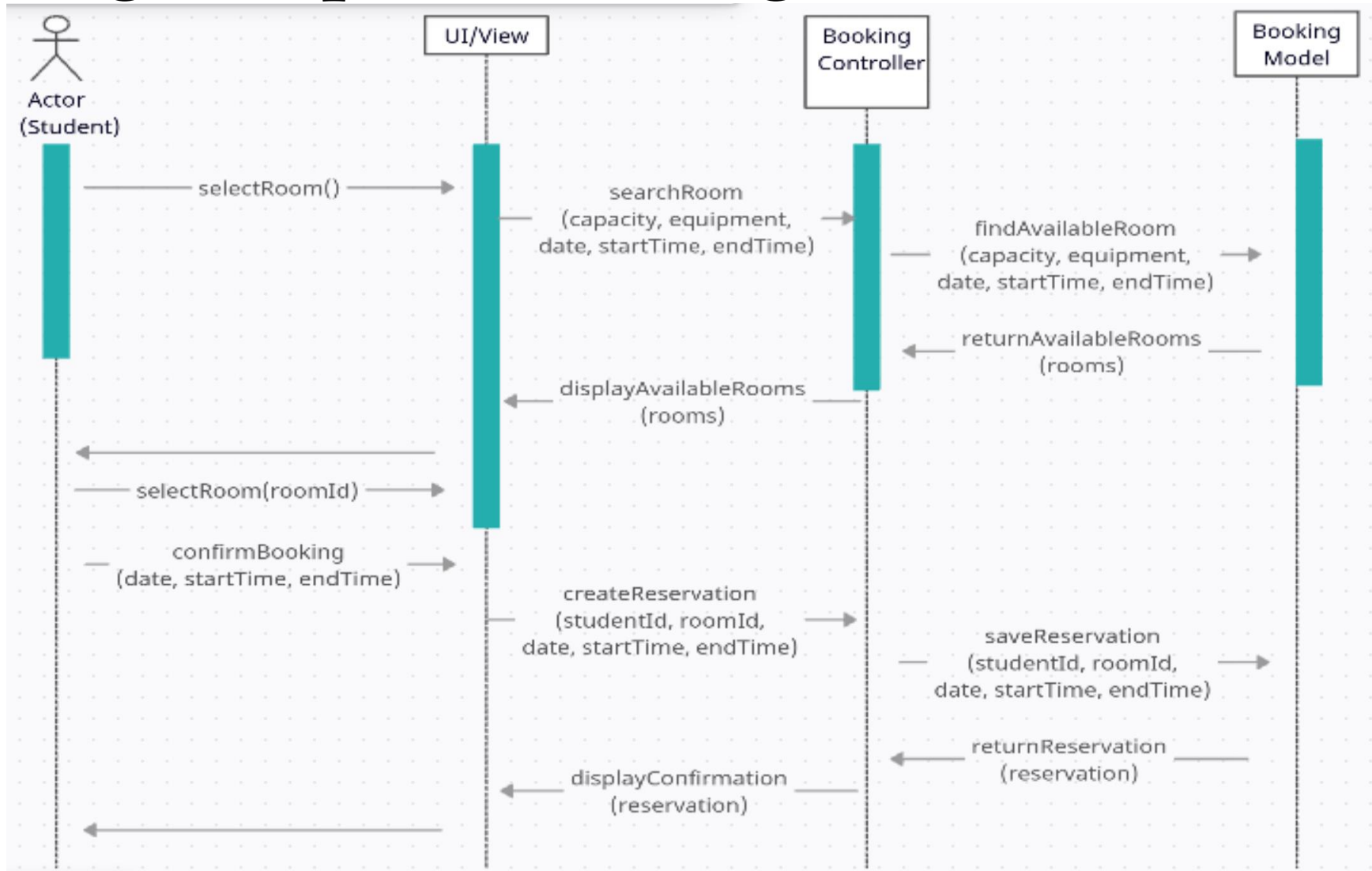
1. The system **shall** display available study rooms within 2 seconds of the user submitting specific filters.
2. The system **shall** encrypt all users' passwords in the database.
3. The system **shall** color code whether a room is available in green and occupied in red.
4. The system **shall** restrict all users to a maximum of 5 hours when booking a study room.
5. The system shall be developed with Spring boot in the backend and React in the frontend.



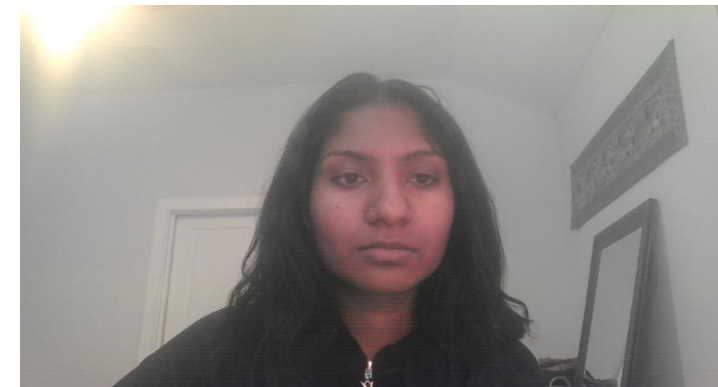
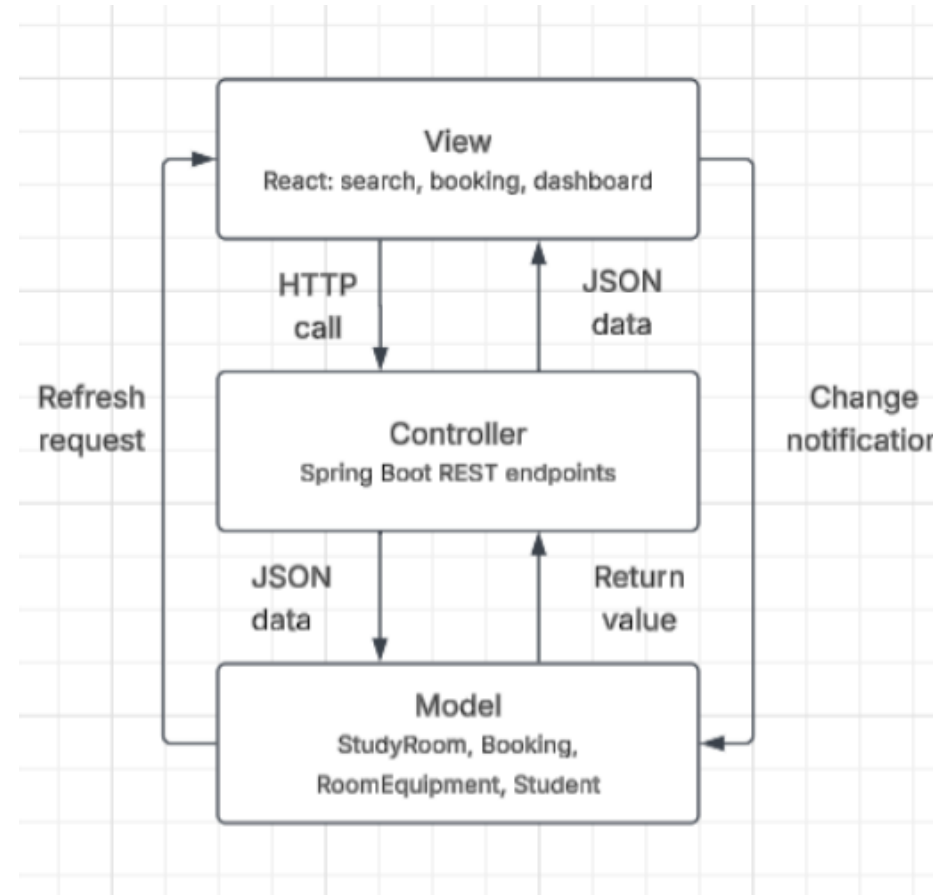
f. Use Case Diagram



g. Sequence Diagram



i. Architectural Design



j. Comparison

- LibCal lets students book rooms without staff help, check in on arrival, and log in with their school account
- Each school sets its own limits on LibCal, such as booking length and how far in advance a room can be reserved
- Our system allows bookings up to a week in advance for a maximum of five hours, more flexible than URI's one day advance limit but still capped like LibCal
- LibCal is a paid product, requiring a license and added costs for extra modules like text reminders or equipment lending
- Our system includes features not standard in most LibCal setups, such as color coded room status (green for open, red for taken) and equipment based filtering
- Google Calendar has been used by some universities, including a pilot program at Fordham, as a low cost alternative to a dedicated system
- The Google Calendar approach worked but required more manual effort from library staff to manage conflicts and confirm booking rules
- Our system automates conflict checking, enforces the five hour limit, and distinguishes student from admin accounts, reducing staff involvement compared to the Google Calendar method



k. True & False Test Case

The system shall restrict all users to a maximum of 5 hours when booking a study room.

True

- Reservation start time: 07/22/26 at 9:30am
- Reservation end time: 07/22/26 at 2:30pm

False

- Reservation start time: 07/22/26 at 9:30am
- Reservation end time: 07/22/26 at 2:31pm

I chose these test cases because they follow to the tee what our requirement stated. Even an extra minute is not valid, and our method should handle it accordingly.

✖	ReservationServiceTest	12 ms
✖	reservationServiceTestInvalid	11 ms
✔	reservationServiceTestValid	1 ms

✖ 1 test failed, 1 passed 2 tests total, 12 ms

/Library/Java/JavaVirtualMachines/jdk-25.jdk/Contents/Home/bin/java ...

java.lang.AssertionError: Here is reservation test c



Prototype

- School only authentication login
- Available for students and administrators

StudySpace
University of Texas at Dallas

Sign in
Use your UTD NetID credentials to continue.

UNIVERSITY EMAIL
netid@utdallas.edu

PASSWORD
UTD password

Sign In

Demo accounts
Student demo Admin demo

Access restricted to enrolled students and university staff.



Prototype

- Student Features
 - Room availability per day and search feature for rooms with filters
 - Amenities provided in each room
 - Number of slots available and max capacity
 - View noise level
 - Availability to check account for current bookings

StudySpace

Browse Rooms

My Bookings

Alex Rivera
axr220089

AR

Reserve a Study Room

Rooms available up to 7 days in advance · 4-hour maximum per session · One active booking per time slot.

Today Tomorrow Mon, Jul 27 Tue, Jul 28 Wed, Jul 29 Thu, Jul 30 Fri, Jul 31

Q Search by room name, building, code, or equipment...

Filters

Whitmore Library · Fl. 2 WL-204

Quiet Available

4

11/13 slots free

Whiteboard Projector Outlets +1 more

Harrington Hall · Fl. 3 HH-3A

Available

8

12/13 slots free

Dual Whiteboards 75" Display Video Conferencing +1 more

Sciences Center · Fl. 1 SC-110

Available

12

13/13 slots free

Projector Whiteboard Outlets +1 more

Whitmore Library · Fl. 1 WL-007

Quiet Available

2

13/13 slots free

Outlets Natural Light Standing Desk

Union Tower · Fl. 5 UT-5B

Available

6

13/13 slots free

Coffee Station Whiteboard Outlets +1 more

Sciences Center · Fl. 3 SC-301

Available

10

Smart Board Outlets +1 more

Available

Booked

Sessions

Prototype

- Student Features
 - Time slots available and duration of session needed
 - Confirmation of reservation
 - Location and equipment available for reservation
 - Current room bookings for account
 - Ability to cancel a reservation

WL-204
Reading Room 204
Whitmore Library · Floor 2 · Up to 4 people

📅 Saturday, July 25, 2026

EQUIPMENT

☐ Whiteboard ☐ Projector ☐ Outlets ☒ Natural Light

START TIME

8:00 AM	9:00 AM	10:00 AM	11:00 AM	12:00 PM
1:00 PM	2:00 PM	3:00 PM	4:00 PM	5:00 PM
6:00 PM	7:00 PM	8:00 PM		

DURATION (max 4 hours)

☒ 1h ☐ 2h ☐ 3h ☐ 4h

Select a valid start time to continue

WL-204
Reading Room 204
Whitmore Library · Floor 2 · Up to 4 people

📅 Saturday, July 25, 2026

EQUIPMENT

☐ Whiteboard ☐ Projector ☐ Outlets ☒ Natural Light

START TIME

8:00 AM	9:00 AM	10:00 AM	11:00 AM	12:00 PM
1:00 PM	2:00 PM	3:00 PM	4:00 PM	5:00 PM
6:00 PM	7:00 PM	8:00 PM		

DURATION (max 4 hours)

☐ 1h ☒ 2h ☐ 3h ☐ 4h

Session 2:00 PM → 4:00 PM · 2h

Confirm Reservation · 2:00 PM – 4:00 PM

StudySpace Browse Rooms My Bookings 1

Alex Rivera
axr220089 AR

My Bookings

1 active reservation. You may not have overlapping time slots.

ROOM
 Reading Room 204 Whitmore Library · WL-204

 1 Reservations	 2h Hours Booked	 1 Bookings
---	--	---



Prototype

- Admin Features
 - View all reservations
 - View active bookings
 - View total bookings for the day
 - View student information for bookings
 - Cancel reservations for students
 - Edit room information and description
 - Change room capacity and location

StudySpace

[Browse Rooms](#)

[My Bookings](#)

[Admin](#)

Dr. Sandra Lee

sxl190001

DS

Admin Panel

6

Total Rooms

3

Active Bookings

2

Students Today

All Reservations

Room Management

RESERVATION	DATE / TIME	STATUS
Reading Room 204 Maria Chen • mxc220045@utdallas.edu	Jul 25, 2026 9:00 AM – 11:00 AM	confirmed Cancel
Seminar Room 110 Maria Chen • mxc220045@utdallas.edu	Jul 26, 2026 2:00 PM – 5:00 PM	confirmed Cancel
Collaboration Suite 3A James Okafor • jxo210012@utdallas.edu	Jul 25, 2026 1:00 PM – 2:00 PM	confirmed Cancel

All Reservations

Room Management

WL-204

Reading Room 204

Whitmore Library • Floor 2 • Capacity 4 • 4 amenities

HH-3A

Collaboration Suite 3A

Harrington Hall • Floor 3 • Capacity 8 • 4 amenities

SC-110

Seminar Room 110

Sciences Center • Floor 1 • Capacity 12 • 4 amenities

WL-007

Focus Booth 07

Whitmore Library • Floor 1 • Capacity 2 • 3 amenities

UT-5B

Graduate Lounge B

Union Tower • Floor 5 • Capacity 6 • 4 amenities

SC-301

Innovation Lab 301

Sciences Center • Floor 3 • Capacity 10 • 4 amenities

Edit Room

ROOM NAME

Reading Room 204

BUILDING

Whitmore Library

ROOM CODE

WL-204

DESCRIPTION

Sunlit corner room with panoramic views of the quad. Ideal

CAPACITY

4

FLOOR

2

☒ Quiet zone

Cancel

Save Changes



m. Challenges, Lessons Learned, & Conclusion

- Deciding on an architecture took discussion, since layered and client-server designs were also considered before settling on MVC
- Coordinating task delegation across five team members required clear division of responsibilities, covering scheduling, cost estimation, testing, and research
- Setting firm rules for the reservation system, such as the five hour time limit, took care to translate into testing logic that correctly separates valid and invalid cases
- Lesson learned: settling on the architecture early made it easier for each team member to work independently without overlapping others' progress
- Lesson learned: comparing the design against existing systems like LibCal and Google Calendar helped clarify which features were worth building versus which added unnecessary complexity
- Conclusion: the project met its original goals, with the main change being the architecture choice rather than a shift in scope
- Conclusion: the system offers a middle ground between a costly established tool like LibCal and a low cost but staff heavy option like Google Calendar

